Course: [BAS102] Engineering Mathematics - I

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: UNIT SUMMARY**

Module / Unit: Unit 1 **Faculty Author: Dr. P. K. Mishra (Professor, D**

Comprehensive Formula Sheet: Matrices, Differential & Vector Calculus

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1. Matrices & Linear Algebra Formulas

- Matrix Rank: Determinantal rank and Echelon row reduction criteria.
- Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation |A - lambda I| = 0.
- Eigenvalues & Eigenvectors: Diagonalization condition (A = P D P^-1).

2. Differential & Vector Calculus Master Formulas

- Leibnitz Theorem for nth derivative of products of two functions u and v.
- Euler's Theorem for Homogeneous Functions: x (dz/dx) + y (dz/dy) = n * z.
- Vector Differential Operators: Gradient (grad f), Divergence (div V), Curl (curl V).